



Chapter 5- Bootable USB, Windows Installation & Virtualization



Creating Bootable USB using Rufus



What is Rufus?



Rufus is a tool used to **create bootable USB drives**



Why we use it?

- Install Windows OS 
- Repair system 
- Boot without OS



Steps to Create Bootable USB



Step 1: Open Rufus



Right-click → **Run as Administrator**



Step 2: Insert Pendrive



Plug USB into laptop



It should be **detected in Rufus**



Step 3: Select ISO File



Select previously downloaded



Windows 11 ISO file

◆ Step 4: Select Partition Scheme (VERY IMPORTANT)

Option Use

MBR For Legacy BIOS

GPT For UEFI

⚠ Important Rule (Data Safety)

👉 If system already installed in **MBR** → select **MBR again**

👉 If you change MBR → GPT → **data can be lost/corrupted**

🧠 How to Identify?

Boot Type Indicator

MBR Straight loading line

GPT Spinning circle

◆ Step 5: Start Process

👉 Click **Start**

◆ Step 6: Customize Windows Installation

👉 Select:

✓ Create Local Account with Username

👉 Click OK

⚠ Warning:

👉 Rufus will say:

“USB data will be deleted”

👉 Click **OK**

🕒 **Result:**

👉 Bootable USB ready for OS installation

🔒 **Hidden Boot ISO (Advanced Concept)**

★ **What is Hidden Boot ISO?**

👉 Special tool used for:

- Password recovery 🗝️
 - Data recovery 💾
 - Bypass login
-

⚠️ **Real Scenario:**

👉 Laptop comes with password lock

Normal Method:

- Format C drive → data lost
-

💡 **Smart IT Support Method:**

👉 Use hidden boot ISO:

- Crack password
 - Access data safely
-

Example:

- 👉 C drive locked → but D/E data needed
 - 👉 Use tool → recover access without data loss
-



Windows OS Installation (Step-by-Step)



Complete Installation Flow

◆ Step 1: Create Bootable USB ✓

◆ Step 2: Enter BIOS

- 👉 Change Boot Order:
 - USB → First priority
-

◆ Step 3: Boot from USB

- 👉 Screen shows:
"Press any key to boot from CD/DVD"

- 👉 Press any key 
-

⚠ If not pressed:

- 👉 System boots into old Windows
-

◆ Step 4: Installation Screen

- Select Language 
 - Click **Install Now**
-

◆ Step 5: License Agreement

✓ Accept → Next

◆ Step 6: Installation Type

👉 Select:

✓ Custom (New Installation)



Disk Partitioning During Installation



Important Screen:



Shows:

- Drive 0 Partition 1
 - Partition 2, 3, 4
-



System Partitions:

- 100 MB
- 16 MB
- 500+ MB



These are created automatically



Clean Installation Steps:

Step 1:



Delete ALL partitions



Result:

- Entire disk becomes **Unallocated**

Step 2:

👉 Click **New Partition**

👉 Example:

- 250 GB → C Drive
 - 250 GB → D Drive
-

Step 3:

👉 Format partitions

★ Why Format?

👉 Converts file system to:

✓ NTFS(**New Technology File System**)

(Windows supported)

💡 Concept:

- Old format → FAT
 - New format → NTFS
-

◆ Step 4:

👉 Click **Next**

🕒 Installation Process:

👉 Files copied from USB → C Drive

👉 Time: ~25–30 minutes

Post Installation Setup

Steps:

1. Select Region 
 2. Select Keyboard 
 3. Add extra keyboard (optional)
 4. Connect Wi-Fi 
-

Important:

-  In Windows 11:
 - ✓ Internet connection required
 -  In Windows 10:
 - ✓ Can skip
-

Microsoft Account Screen

-  Select:
 - ✓ Continue with Limited Setup
-

Privacy Settings:

-  Disable unnecessary options
-

Final Result:

-  Windows installed successfully
-

Troubleshooting Scenarios (VERY IMPORTANT)

USB Not Booting

Cause:

- USB boot disabled

Fix:

 Enable **USB Boot in BIOS**

OS Not Installing

Causes:

- Wrong partition format
 - Disk not cleaned
-

Data Loss Risk

 Wrong partition scheme selected

Virtual Machine

What is Virtual Machine?

 Software that allows you to run OS inside another OS

Example:

-  Run Windows inside Windows
 -  Practice without affecting real system
-

Tools for Virtual Machine

1 VMware Workstation

- 👉 Most popular tool
 - 👉 Used in companies
-

2 Oracle VirtualBox

- 👉 Free tool
 - 👉 Good for beginners
-

Basic Installation Steps (VMware / VirtualBox)

Step 1:

- 👉 Download & install software
-

Step 2:

- 👉 Click **Create New Virtual Machine**
-

Step 3:

- 👉 Select ISO file (Windows)
-

Step 4:

- 👉 Allocate:
 - RAM (4GB minimum)
 - Storage (50GB recommended)
-

Step 5:

👉 Start VM → Install OS

🎯 Why Virtual Machine is Important?

👉 Practice:

- OS installation
- Virus testing
- Networking

🚀 Real IT Support Mindset

- ✓ Always take backup before format
 - ✓ Check partition type (MBR/GPT)
 - ✓ Verify boot order
 - ✓ Use tools smartly (Rufus, VM)
 - ✓ Avoid direct data loss
-

🎲 Final Understanding

- 👉 Rufus → Bootable USB
 - 👉 BIOS → Boot control
 - 👉 Partition → Disk structure
 - 👉 OS Install → Practical skill
-

Chapter (SUMMARY)

Bootable USB (Rufus Tool)

 **Rufus** = Tool to create bootable USB 

Uses:

- Install Windows 
- Repair OS 
- Boot system without OS

Bootable USB Process

1. Open Rufus (Admin mode) 
2. Insert USB 
3. Select ISO file 
4. Choose Partition Scheme 
 - MBR → Legacy
 - GPT → UEFI
5. Click Start 

Important Concept

 Wrong partition selection = Data loss 

Hidden Boot ISO (Advanced)

 Used for:

- Password recovery 
- Data recovery 
- Bypass login

 Smart IT Support:

 Data save karke access lena (format nahi karna)

Windows Installation (Flow)

 **Steps:**

1. Boot from USB 
 2. Install Windows
 3. Select Custom install
 4. Delete partitions
 5. Create new partitions
 6. Format (NTFS)
 7. Install OS
-

Partitioning Concept

- C Drive → OS
- D Drive → Data

 Format = File system set (NTFS)

Post Installation

- Region 
- Keyboard 

- Wi-Fi 🌐
 - Account setup 👤
-

Virtual Machine

👉 VM = OS inside OS

Tools:

- VMware
- VirtualBox

👉 Use:

- Practice
 - Testing
 - Safe environment
-

CONCLUSION

👉 Core understanding:

- Rufus = Bootable USB
- BIOS = Boot control
- Partition = Disk management
- VM = Safe practice

👉 Final line:

“A real IT Support engineer never formats directly — first tries to save data”



Q & A

◆ BOOTABLE USB QUESTIONS

? Q1: What is Rufus?

👉 Answer:

Rufus is a tool used to create bootable USB drives. It helps in installing operating systems or repairing systems.

? Q2: Why do we use bootable USB?

👉 Answer:

- Install OS
 - Repair system
 - Boot system without installed OS
-

? Q3: What is ISO file?

👉 Answer:

An ISO file is an image file that contains complete OS installation data.

? Q4: Difference between MBR and GPT in Rufus?

👉 Answer:

- MBR → For Legacy BIOS
- GPT → For UEFI

👉 Choosing wrong option can cause installation failure.

◆ WINDOWS INSTALLATION QUESTIONS

? **Q5: What are steps to install Windows?**

👉 **Answer:**

1. Create bootable USB
 2. Enter BIOS
 3. Set USB boot priority
 4. Boot from USB
 5. Select Custom install
 6. Partition disk
 7. Install OS
-

? **Q6: What is NTFS?**

👉 **Answer:**

NTFS (New Technology File System) is a file system used by Windows to store and manage files efficiently.

? **Q7: Why do we format drive?**

👉 **Answer:**

Formatting prepares the disk by setting file system and removing old data.

◆ PARTITION QUESTIONS

? **Q8: Why delete partitions before install?**

👉 **Answer:**

To remove old OS and create clean installation environment.

? Q9: What happens if we install OS without formatting?

👉 Answer:

Old files remain and system may become slow or unstable.

◆ TROUBLESHOOTING QUESTIONS (VERY IMPORTANT)

🔌 USB NOT BOOTING

? Q10: USB not booting, what will you do?

👉 Answer:

1. Check BIOS settings
 2. Enable USB boot
 3. Check boot order
 4. Recreate bootable USB
-

✗ OS NOT INSTALLING

? Q11: Windows not installing?

👉 Answer:

- Wrong partition type
- Disk not formatted
- Corrupt ISO

👉 Fix:

- Convert disk (MBR/GPT)
 - Recreate USB
-

⚠ DATA LOSS SCENARIO

❓ Q12: How will you avoid data loss?

👉 Answer:

- Take backup
 - Use recovery tools
 - Do not format directly
-

◆ ADVANCED QUESTIONS

❓ Q13: What is Hidden Boot ISO?

👉 Answer:

A special bootable tool used to recover passwords, access locked systems, and recover data without formatting.

❓ Q14: When do you use Hidden Boot ISO?

👉 Answer:

When system is locked and important data needs to be recovered without deleting it.

◆ VIRTUAL MACHINE QUESTIONS

❓ Q15: What is Virtual Machine?

👉 Answer:

A virtual machine is software that allows running an operating system inside another OS.

? Q16: Why VM is used?

👉 Answer:

- Practice OS installation
 - Test software
 - Safe environment
-

? Q17: VMware vs VirtualBox?

👉 Answer:

- VMware → Professional use
 - VirtualBox → Free & beginner-friendly
-

◆ REAL JOB SCENARIOS

? Q18: Laptop comes with password lock, what will you do?

👉 Answer:

- Try recovery tools
 - Use hidden boot ISO
 - Avoid formatting if data important
-

? Q19: System boots into old OS even after USB inserted?

👉 Answer:

👉 Boot priority not set

👉 Fix:

- Set USB as first boot device
-

? Q20: Disk not showing during installation?

👉 Answer:

- Driver missing
- Disk issue

👉 Fix:

- Load drivers
 - Check BIOS
-

◆ INTERVIEW IMPRESSION QUESTIONS

? Q21: How do you perform clean OS installation?

👉 Answer:

- Delete partitions
 - Create new partitions
 - Format with NTFS
 - Install OS
-

? Q22: What precautions you take before OS install?

👉 Answer:

- Backup data
 - Check partition type
 - Verify ISO
 - Check BIOS mode
-